

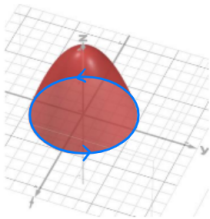
Strokes therom

Strokes therom allows you to return a surface integral into a line integral

but orientation matters when you are applying this therom you need to go counterclockwise around the normal vector when you are going counterclockwise you are assuming that the normal vector is pointing directly outward so you can use youre right hand rule where your thumb is the direction of the normal vector and where ever your Fingers curl is the direction but go counter clock wise unless you are told the normal vector Points anywhere else

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}(\vec{F}) \cdot d\vec{S}$$

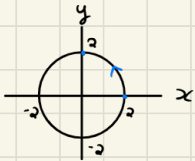
1. Use Stroke's Theorem to evaluate $\iint_S \text{curl}(\vec{F}) \cdot d\vec{S}$ where $\vec{F}(x, y, z) = x^2 \sin z \vec{i} + y^2 \vec{j} + xy \vec{k}$, and S is part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy -plane, oriented upwards.



its on the xy -Plane so the z -axis is zero

We set the $z=0$ because as stated above its on the xy -Plane

$$z = 4 - x^2 - y^2 \Rightarrow 0 = 4 - x^2 - y^2 \Rightarrow x^2 + y^2 = 4$$



$\vec{r}(t)$:

$$x = 2 \cos t$$

$$y = 2 \sin t$$

$$0 \leq t \leq 2\pi$$

we can check if this goes counterclockwise

t	x	y
0	2	0
$\pi/2$	0	2

you can look at the graph and see the movement

$$\vec{F} = x^2 \sin z \vec{i} + y^2 \vec{j} + xyz \vec{k}$$

$$\vec{r}(t) = 2 \cos t \vec{i} + 2 \sin t \vec{j} + 0 \vec{k}$$

$$\vec{F}(\vec{r}(t)) = (2 \cos t)^2 \sin(0) \vec{i} + (2 \sin t)^2 + (2 \cos t)(2 \sin t) \vec{k}$$

$$\rightarrow \vec{F}(\vec{r}(t)) = 0 \vec{i} + 4 \sin^2 t \vec{j} + 4 \cos t \sin t \vec{k}$$

$$\vec{r}'(t) = -2 \sin t \vec{i} + 2 \cos t \vec{j} + 0 \vec{k}$$

$$\vec{F}(\vec{r}) \cdot \vec{r}' = 8 \sin^2(t) \cos(t)$$

$$\int_0^{2\pi} 8 \sin^2(t) \cos(t) dt$$

$$\int_0^{2\pi} 8(u)^2 \cos t \frac{du}{\cos t}$$

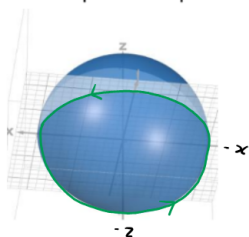
$$u = \sin t$$

$$du = \cos t dt$$

$$\frac{du}{\cos t} = dt$$

$$\left[\frac{8}{3} (\sin t)^3 \right]_0^{2\pi} \Rightarrow \frac{8}{3} \sin^3(2\pi) - \frac{8}{3} \sin^3(0) = \boxed{0}$$

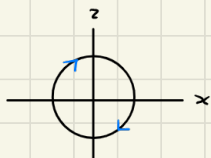
2. Use Stoke's Theorem to evaluate $\iint_S \text{curl}(\vec{F}) \cdot d\vec{S}$ where $\vec{F}(x, y, z) = ze^y \vec{i} + x \cos y \vec{j} + xz \sin y \vec{k}$, and S is part of the paraboloid $x^2 + y^2 + z^2 = 4, y \geq 0$ oriented in the direction of the positive y-axis.



the y-axis is parameterized to zero

as stated above we parametrize by setting $y=0$

$$x^2 + y^2 + z^2 = 4 \Rightarrow x^2 + z^2 = 4$$



t	x	y
0	2	0
$\pi/2$	0	-2

if we look at the movement from the 3d graph we began at the $-x$ the go to $+z$ we are going in the counter clock wise direction if we now go from the $-x$ to $+z$ we will see we are going clockwise

$\vec{r}(t)$:

$$x = 2 \cos(-t) \Rightarrow x = 2 \cos(t)$$

$$z = 2 \sin(-t) \Rightarrow z = -2 \sin(t)$$

$$0 \leq t \leq 2\pi$$

} we include negatives
because we went clock
wise and we need to go
opposite

$$\cos(-t) = \cos(t)$$

$$\sin(-t) = -\sin(t)$$

$$\vec{F} = 2e^y \vec{i} + x \cos y \vec{j} + xz \sin y \vec{k}$$

$$\vec{r}(t) = 2 \cos(t) \vec{i} + 0 \vec{j} - 2 \sin(t) \vec{k}$$

} $\vec{F}(\vec{r}) = (-2 \sin(t)) e^0 \vec{i} + (2 \cos(t)) \cos(0) \vec{j} + (2 \cos(t)) \sin(0) \vec{k}$
 $\vec{F}(\vec{r}) = -2 \sin(t) \vec{i} + 2 \cos(t) \vec{j} + 0 \vec{k}$
 $\vec{r}'(t) = -2 \sin(t) \vec{i} + 0 \vec{j} - 2 \cos(t) \vec{k}$

$$\vec{F}(\vec{r}) \cdot \vec{r}' = 4 \sin^2(t)$$

$$\int_0^{2\pi} 4 \sin^2(t) dt \Rightarrow 4 \int_0^{2\pi} \frac{1}{2} - \frac{1}{2} \cos(2t) dt \Rightarrow 4 \int_0^{2\pi} \frac{1}{2} dt - \frac{1}{2} \int_0^{2\pi} \cos(2t) dt$$

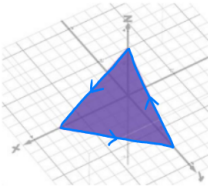
$$u = 2t \\ du = 2 dt \\ \frac{du}{2} = dt$$

$$4 \int_0^{2\pi} \frac{1}{2} dt - \frac{1}{4} \int_0^{2\pi} \cos(u) du \Rightarrow 4 \left[\frac{1}{2} t - \sin(2t) \right]_0^{2\pi}$$

$$4 \left[\frac{1}{2} (2\pi) - \sin(4\pi) \right] - 4 \left[0 - \sin(0) \right]$$

$$(4\pi - 0) - 0 = \boxed{4\pi}$$

3. Use Stoke's Theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is oriented **counter clockwise** as viewed from above, $\vec{F}(x, y, z) = (x + y^2)\vec{i} + (y + z^2)\vec{j} + (z + x^2)\vec{k}$, C is a triangle with vertices $(3, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 3)$ (or C is the boundary of the plane $x + y + z = 3$ in the **first quadrant**).



you are not given the curl you are given $\vec{F}$

$$\int_C \vec{F} \cdot d\vec{r} = \iint \text{curl}(\vec{F}) \cdot d\vec{s}$$

$$= \iint \left(-P \frac{\partial q}{\partial x} - Q \frac{\partial q}{\partial y} + R \right) dA$$

$$\text{curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x+y^2) & (y+z^2) & (z+x^2) \end{vmatrix}$$

$$\vec{i} \left[\frac{\partial}{\partial y} (z+x^2) - \frac{\partial}{\partial z} (y+z^2) \right] - \vec{j} \left[\frac{\partial}{\partial x} (z+x^2) - \frac{\partial}{\partial z} (x+y^2) \right] + \vec{k} \left[\frac{\partial}{\partial x} (y+z^2) - \frac{\partial}{\partial y} (x+y^2) \right]$$

$$= -2z\vec{i} - 2x\vec{j} - 2y\vec{k}$$

$\downarrow$
P

$\downarrow$
Q

$\downarrow$
R

$$g(x, y) = x + y + z = 3 \Rightarrow z = 3 - x - y$$

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \underbrace{\text{curl}(\vec{F})}_{P, Q, R} \cdot d\vec{s}$$

$$\iint (2x - 2z - 2y) dA$$

* you need to have it in x and y so you will replace z with $z = 3 - x - y$

$$\iint (-2x - 2(3 - x - y) - 2y) dA$$

You were told that the Plane is in the First quadrant so that means $z=0$

$$\begin{aligned}x+y+z &= 3 \\x+y &= 3\end{aligned}$$

$$0 \leq x \leq 3$$

$$0 \leq y \leq -x+3$$

$$x+y=3$$

$$y = -x+3$$

} we isolated for y

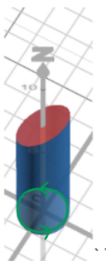


$$\int_0^3 \int_0^{-x+3} (-\cancel{2x} - 6 + \cancel{2x} + \cancel{2y} - \cancel{2y}) dy dx \Rightarrow \int_0^3 \int_0^{-x+3} -6 dy dx$$

$$= \left[-6y \right]_0^{-x+3} \Rightarrow -6(-x+3) = 6x - 18$$

$$\int_0^3 6x - 18 dy \Rightarrow \left[3x^2 - 18x \right]_0^3 \Rightarrow 3(3)^2 - 18(3) = \boxed{-27}$$

4. Use Stoke's Theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is oriented **counter clockwise** as viewed from above, $\vec{F}(x,y,z) = 3y\vec{i} + xz\vec{j} + (x+y)\vec{k}$, C is the curve of intersection of the plane $z = y+7$ and the cylinder $x^2 + y^2 = 1$



you are not given the curl

$$\text{curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3y & xz & (x+y) \end{vmatrix}$$

$$\vec{i} \left[\frac{\partial}{\partial y} (x+y) - \frac{\partial}{\partial z} (xz) \right] - \vec{j} \left[\frac{\partial}{\partial x} (x+y) - \frac{\partial}{\partial z} (3y) \right] + \vec{k} \left[\frac{\partial}{\partial x} (xz) - \frac{\partial}{\partial y} (3y) \right]$$

$$\begin{aligned} (1-x)\vec{i} &- \vec{j} + (2-3)\vec{k} \\ \downarrow & \quad \downarrow \quad \downarrow \\ \text{p} & \quad \text{q} \quad \text{r} \end{aligned}$$

$$g(x,y) \Rightarrow z = y+7$$

$$\frac{\partial g}{\partial x} = 0 \quad \text{and} \quad \frac{\partial g}{\partial y} = 1$$

$$\iint_D \left(-P \frac{\partial g}{\partial x} - Q \frac{\partial g}{\partial y} + R \right) dA$$

$$\iint \left((1-x)(0) + (1) + (z-3) \right) dA$$

$$\iint \left(1 + ((y+7) - 3) \right) dA$$

$$\iint y + 5 \, dA$$

$$\int_0^{2\pi} \int_0^1 (\sin\theta + 5) r \, dr \, d\theta$$

~~*~~ We don't want z in our integral so replace it with $z = y + 7$

We will do this in Polar coordinates:

$$0 \leq \theta \leq 2\pi$$

$$0 \leq r \leq 1$$

$$y = r \sin\theta \Rightarrow (1) \sin\theta$$

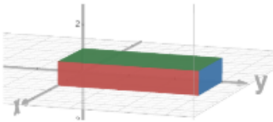
Divergence theorem

the divergence theorem states if you have a surface integral you can return it into a triple integral

$$\underbrace{\iint \vec{F} \cdot d\vec{s}}_{\text{Flux}} = \iiint_E (\nabla \cdot \vec{F}) dV$$

Problem Set

1. Use the Divergence Theorem to calculate the Surface Integral $\iint_S \vec{F} \cdot d\vec{S}$ (Flux of $\vec{F}$ across S) where $\vec{F}(x, y, z) = xye^z \vec{i} + xy^2z^3 \vec{j} - ye^z \vec{k}$, and S is the surface of the box bounded by the coordinate planes and the planes $x = 3$, $y = 6$, and $z = 1$.



$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$$

$$\nabla \cdot \vec{F} = \cancel{ye^z} + 2yxz^3 - \cancel{ye^z} = \nabla \cdot \vec{F} = 2yxz^3$$

When you are dealing with rectangles they are going to be much easier than any other shape

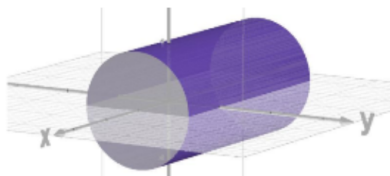
$$0 \leq x \leq 3$$

$$0 \leq y \leq 6$$

$$0 \leq z \leq 1$$

$$\int_0^1 \int_0^6 \int_0^3 2yxz^3 dx dy dz$$

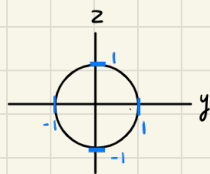
2. Use the Divergence Theorem to calculate the Surface Integral $\iint_S \vec{F} \cdot d\vec{S}$ (Flux of $\vec{F}$ across S) where $\vec{F}(x, y, z) = 3xy^2\vec{i} + xe^z\vec{j} + z^3\vec{k}$, and S is the surface of the solid bounded by the cylinder $y^2 + z^2 = 1$ and the planes $x = -3$, and $x = 1$.



$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E (\nabla \cdot \vec{F}) dV$$

$$\nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial P}{\partial y} + \frac{\partial P}{\partial z}$$

$$\nabla \cdot \vec{F} = 3y^2 + 0 + 3z^2 \Rightarrow 3y^2 + 3z^2$$

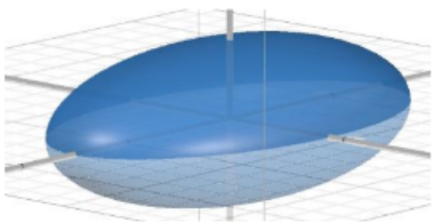


we get this looking at it in 2d
we know 1 is the radius because
 $x^2 + z^2 = 1$ was given to us

$$\begin{aligned} -3 &\leq x \leq 1 \\ -1 &\leq y \leq 1 \\ -\sqrt{1-y^2} &\leq z \leq \sqrt{1-y^2} \end{aligned}$$

$$\int_{-3}^1 \int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} (3y^2 + 3z^2) dz dy dx$$

3. Use the Divergence Theorem to calculate the Surface Integral $\iint_S \vec{F} \cdot d\vec{S}$ (Flux of $\vec{F}$ across S) where $\vec{F}(x, y, z) = xe^y \vec{i} + (z - e^y) \vec{j} - xy \vec{k}$, and S is the surface of the solid bounded by the ellipsoid $x^2 + 3y^2 + 8z^2 = 9$



$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_V (\nabla \cdot \vec{F}) dV$$

$$\nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial P}{\partial y} + \frac{\partial P}{\partial z}$$

$$\nabla \cdot \vec{F} = e^y - e^y + 0 \Rightarrow 0$$

the answer to this will just be zero but you would need to write it like this

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_V 0 = \boxed{0}$$

but how would you do this if you actually had to solve it how would the integral limits look

outer

$$x^2 + 3y^2 + 8z^2 = 9 \quad @ y=0 \text{ and } z=0, \text{ because you want limits for } x$$

$$x^2 = 9 \leadsto x = \pm 3 \quad -3 \leq x \leq 3$$

inner

$$x^2 + 3y^2 + 8z^2 = 9 \longrightarrow 8z^2 = 9 - x^2 - 3y^2 \Rightarrow z = \pm \sqrt{\frac{9 - x^2 - 3y^2}{8}} \quad \sqrt{\frac{9 - x^2 - 3y^2}{8}} \leq z \leq \sqrt{\frac{9 - x^2 - 3y^2}{8}}$$

to find y because you are setting up a triple integral

$z=0$:

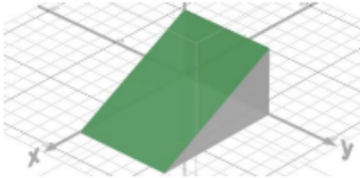
$$x^2 + 3y^2 = 9 \Rightarrow 3y^2 = 9 - x^2 \Rightarrow y = \pm \sqrt{\frac{9 - x^2}{3}}$$

$$-\sqrt{\frac{9 - x^2}{3}} \leq y \leq \sqrt{\frac{9 - x^2}{3}}$$

So setting the integral it would look something like this

$$\int_{-3}^3 \int_{-\sqrt{\frac{9-x^2}{3}}}^{\sqrt{\frac{9-x^2}{3}}} \int_0^{\frac{9-x^2-3y^2}{8}} dz \, dy \, dx$$

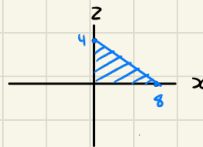
4. Use the Divergence Theorem to calculate the Surface Integral $\iint_S \vec{F} \cdot d\vec{S}$ (Flux of $\vec{F}$ across S) where $\vec{F}(x, y, z) = x^2 z \vec{i} + x z^3 \vec{j} + y \ln(x+1) \vec{k}$, and S is the surface of the solid bounded by the planes $x + 2z = 8$, $y = 6$, $x = 0$, $y = 0$, $z = 0$.



$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_V (\nabla \cdot \vec{F}) \, dV$$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$$

$$\nabla \cdot \vec{F} = 2xz + 0 + 0 \Rightarrow 2xz$$



$$x + 2z = 8$$

$$z = -\frac{x}{2} + 4$$

$$0 \leq x \leq 8$$

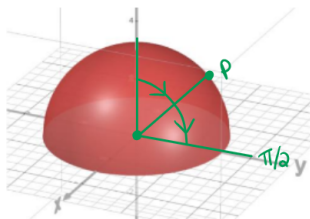
$$0 \leq y \leq 6$$

$$0 \leq z \leq -\frac{x}{2} + 4$$

$$\int_0^8 \int_0^6 \int_0^{-\frac{1}{2}x+4} 2xz \, dz \, dy \, dx$$

5. Use the Divergence Theorem to calculate the Surface Integral $\iint_S \vec{F} \cdot d\vec{S}$ (Flux of $\vec{F}$ across S) where

$\vec{F} = |\vec{r}| \vec{r}$ where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, and S consist of the hemisphere $z = \sqrt{9 - x^2 - y^2}$ and the disk $x^2 + y^2 \leq 9$ in the xy-plane.



Find the magnitude of $\vec{r}$ and then multiply it by $\vec{r}$ that will be your $\vec{F}$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\vec{F} = x\sqrt{x^2 + y^2 + z^2} \vec{i} + y\sqrt{x^2 + y^2 + z^2} \vec{j} + z\sqrt{x^2 + y^2 + z^2} \vec{k}$$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$$

$$\nabla \cdot \vec{F} = \sqrt{x^2 + y^2 + z^2} + \frac{x^2}{\sqrt{x^2 + y^2 + z^2}} + \sqrt{x^2 + y^2 + z^2} + \frac{y^2}{\sqrt{x^2 + y^2 + z^2}} + \sqrt{x^2 + y^2 + z^2} + \frac{z^2}{\sqrt{x^2 + y^2 + z^2}}$$

$$\nabla \cdot \vec{F} = 3\sqrt{x^2 + y^2 + z^2} + \frac{(x^2 + y^2 + z^2)}{\sqrt{x^2 + y^2 + z^2}} \Rightarrow 3\sqrt{x^2 + y^2 + z^2} + \sqrt{x^2 + y^2 + z^2}$$

$$\nabla \cdot \vec{F} = 4\sqrt{x^2 + y^2 + z^2}$$

We would use Spherical coordinates:

$$0 \leq \theta \leq 2\pi$$

$$0 \leq \rho \leq 3$$

$$0 \leq \phi \leq \pi/2$$

$$z^2 = 9 - x^2 - y^2$$

$$x^2 + y^2 + z^2 = 9$$

$$\rho^2 = 9$$

$$\rho = \pm 3$$

$$\int_0^{2\pi} \int_0^3 \int_0^{\pi/2} 4\sqrt{x^2+y^2+z^2} (p^2 \sin \phi \, d\phi \, dp \, d\theta)$$

$$\int_0^{2\pi} \int_0^3 \int_0^{\pi/2} 4r^3 \sin \phi \, d\phi \, dp \, d\theta$$

$$\int_0^{2\pi} \int_0^3 \int_0^{\pi/2} 4p^3 \sin \phi \, d\phi \, dp \, d\theta$$